

FRAMEWORK SERIES

The Performance Engine

by design

Sustained performance comes from the environment that surrounds the work. This piece sets out a model for building that environment deliberately, by growing the capacity people bring to the work and removing the interference that stops it showing up, then running a cycle that keeps the system learning.

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INTRODUCTION

The heroics problem

When results dip or a demanding period arrives, the instinctive response in many high performance environments is to ask people for more. More hours and more intensity get requested from staff and athletes who are usually already giving a great deal. For a short period that often works, because talented and committed people can cover for a surprising amount of structural weakness when they care about the outcome.

The trouble lies in what the extra effort hides. Across two decades inside professional rugby union and Major League Baseball player development, I have seen environments keep winning while they accumulate fatigue, rushed decisions and unclear ownership alongside a growing dependence on the few people who hold everything together. On the scoreboard those environments look healthy, while underneath they are becoming fragile in ways that only surface when one of those people leaves, gets injured or simply runs out of energy.

The Performance Engine is a systems model built to explain that pattern and to give leaders a practical way out of it. It treats performance as an output of the environment that surrounds the work, which turns the most useful leadership question into a design question. What has to be built and what has to be removed so that good performance becomes easier to produce?

The blueprint is blunt about one diagnosis in particular. A system that needs regular heroics to survive is a broken system, however impressive those heroics look from the outside and however grateful everyone is for the people providing them.

THE CORE RELATIONSHIP

Performance

= Capacity

(Additions & Multipliers)

– Interference

(Subtractions & Dividers)

Performance improves when an organisation deliberately increases useful capacity while reducing the interference that prevents that capacity from being expressed.

If your system needs heroics to survive, it is broken.

THE PERFORMANCE
ENGINE BLUEPRINT

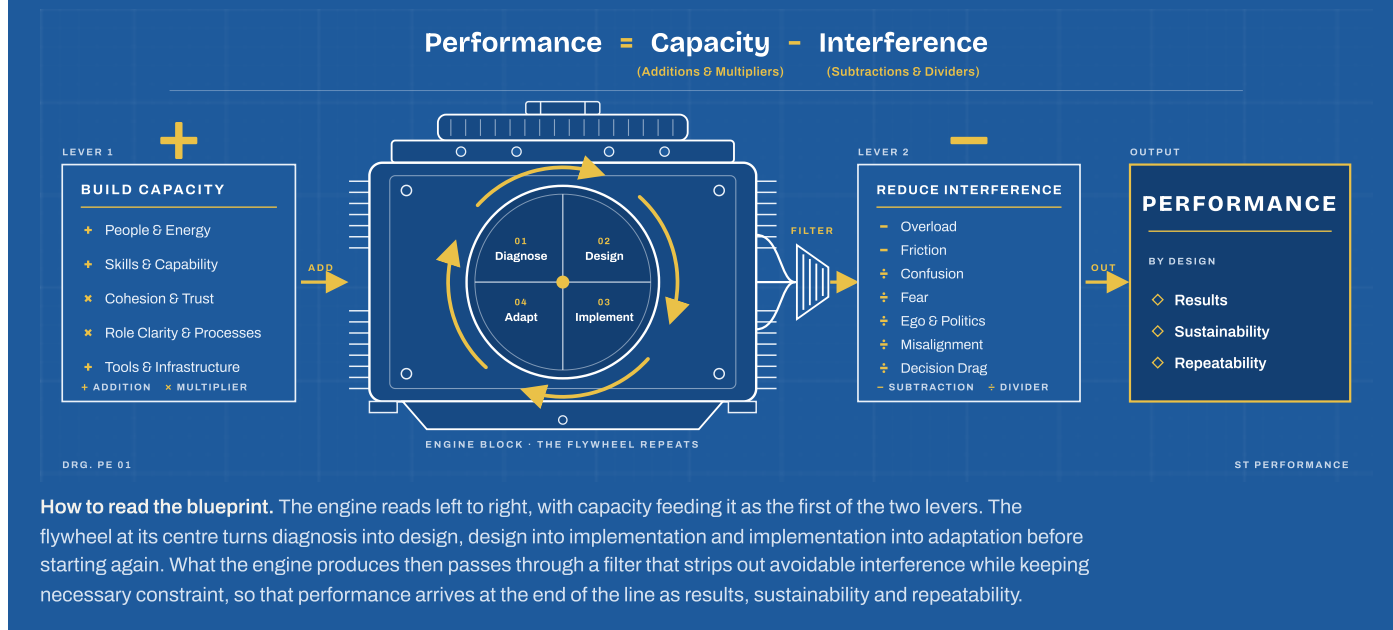
THE DESIGN QUESTION

What must we build and what must we remove so that good performance becomes easier to produce, again and again, without relying on the same few people to rescue it?

The engine and its equation

FIG. 1 THE PERFORMANCE ENGINE BLUEPRINT

ST PERFORMANCE SYSTEM ARCHITECTURE



The model rests on a single relationship in which performance equals capacity minus interference. Capacity is everything the system can bring to the work when it is functioning well. It covers the energy and skill of the people involved along with the quality of their relationships, the clarity of their roles and the tools they have to work with. Interference is anything that consumes that capacity without creating equivalent value, such as overload, friction, confusion or the slow drag of decisions that never quite get made.

Performance is whatever remains once interference has taken its share. Two groups with similar talent can therefore produce very different results, because one has built an environment that lets its capacity reach the work while the other loses a large part of it to noise before anything useful happens.

The engine metaphor was chosen with care. An engine can contain excellent individual components and still perform poorly if those components are misaligned, overloaded or unable to transfer energy to where it is needed. Organisations behave in much the same way, which is why a collection of well intentioned initiatives rarely adds up to better performance unless someone is paying attention to how the parts connect.

The blueprint gives those three outputs equal weight, because results without sustainability produce a short lived peak while sustainability without meaningful results amounts to little more than comfort. Repeatability is what turns an isolated success into a capability the organisation can rely on season after season.

THE OUTPUT

What the engine is built to produce

- ◇ **Results.** The outcomes the environment exists to deliver, measured honestly.
- ◇ **Sustainability.** Results produced without exhausting the people who produce them.
- ◇ **Repeatability.** Success the system can reproduce next week without heroics.

What capacity and interference are made of

Each side of the equation has two parts. The difference between them explains a great deal about why effort alone so rarely fixes a struggling environment. Capacity is made up of additions and multipliers. An addition puts a fixed amount of capacity into the system, such as a new hire, a new skill, protected recovery time or a better tool. A multiplier raises the value of everything already there, which is what trust, clarity, shared standards and good leadership do for a group whose talent has not changed at all.

Interference mirrors that structure with subtractions and dividers. A subtraction takes a fixed amount away, whether it is a meeting that produces nothing, data entered twice or a travel day planned badly. A divider cuts into every unit of capacity at once, which is why unclear priorities, fear, politics and decisions that never land do so much more damage than their visibility suggests.

All four types show up in resources and logistics, in the physical environment and the culture and in the systems, people and processes that carry the work, so the same category can sit on either side of the equation depending on how it has been designed.

Applied to the seven forms of interference, the test is revealing. The doubling test in the panel alongside makes the pattern clear. Overload and friction mostly subtract, while the remaining five forms mostly divide, which goes a long way to explaining why working harder rarely fixes them. Extra effort is itself an addition, which means it gets divided along with everything else the moment it enters a divided system.

The practical order for a leader follows from that. Remove dividers first, protect and build multipliers next, cut the subtractions that remain and add capacity last, even though adding is where many organisations instinctively start.

Where the four types fall across the organisation

The examples are drawn from high performance environments. Each category can add or multiply capacity when it is well designed, or subtract and divide it when it is not.

CATEGORY	+ ADDITION	× MULTIPLIER	- SUBTRACTION	÷ DIVIDER
Resources	Extra staff, budget or equipment	Resources placed where the real constraint is	Budget spent on low value work	Resources spread thinly across competing priorities
Environment	Recovery spaces and better facilities	A setting where the standard is visible every day	Noise, poor facilities and time lost between sites	An environment that rewards looking busy
Culture	New rituals, recognition and onboarding	Trust, candour and shared standards	Time lost to unresolved conflict	Fear, ego and politics
Logistics	Better scheduling and travel support	Planning that removes friction before it arrives	Travel days, waiting and double handling	Constant rescheduling and late changes
Systems	A new data platform or reporting tool	One version of the truth that everyone uses	Manual reporting and duplicated entry	Systems that contradict each other
People	Hires, development and specialist support	Leaders who lift the people around them	Absence, injury and turnover	Misaligned people who dilute the effort of others
Processes	A documented workflow	Clear decision rights and ownership	Unnecessary approvals and handoffs	Decisions revisited again and again

THE FOUR TYPES

+ Additions. Put a fixed amount of capacity into the system.

× Multipliers. Raise the value of everything already there.

- Subtractions. Take a fixed amount of capacity away.

÷ Dividers. Cut into every unit of capacity at once.

THE DOUBLING TEST

Ask whether an effect would double if everything else in the environment doubled. A proportional effect marks a multiplier or a divider, while a fixed effect marks an addition or a subtraction.

Where the equation comes from

The equation will feel familiar to anyone who has read Timothy Gallwey. His work on the inner game, which began with *The Inner Game of Tennis* in 1974, is often summarised as performance equals potential minus interference. His central argument was that the largest obstacles to a player's performance are usually generated inside the player's own head. The Performance Engine takes that insight one level out to the environment around the individual, where the interference that matters most in a professional organisation is often built into the schedule, the meeting structure and the unspoken rules about who is allowed to disagree with whom.

The idea that the system shapes the result has a long history in management as well. In *Out of the Crisis*, W. Edwards Deming estimated that around 94 percent of the problems and possibilities for improvement in an organisation belong to the system, leaving only a small share attributable to the individuals working inside it. He placed responsibility for that system squarely with management, which is where the Performance Engine places it too. In sport the same logic applies whenever a head coach is replaced while the structure around the role stays exactly as it was. The new appointment then inherits the same interference that limited the last one.

Research on teams points in the same direction. Ruth Wageman and J. Richard Hackman's work on senior leadership teams produced a finding that is now widely cited in team coaching: roughly sixty percent of the difference in how well a team performs comes from how it has been designed, around thirty percent from how it was launched and only about ten percent from the coaching it receives once it is underway. That is a sobering distribution for leaders, because so much leadership energy goes into the final ten percent while the conditions that account for most of the outcome are left to chance.

Occupational psychology offers a fourth lens. The job demands and resources model developed by Arnold Bakker and Evangelia Demerouti shows that exhaustion and disengagement grow when the demands placed on people outstrip the resources available to meet them, while engagement grows when resources are strong enough to turn demand into challenge.

The Performance Engine borrows its equation openly from this lineage. What it adds is an operational structure a leader can actually run, with named components of capacity, named forms of interference and a repeating cycle for working on both.

For a leader the practical implication of this lineage is uncomfortable and useful in equal measure. If most of the variation in performance sits in the design of the environment, most of the room for improvement sits there too, which makes the time spent examining that design some of the most valuable time a leadership group can spend.

SOURCES

- ◇ Gallwey, T. (1974). *The Inner Game of Tennis*. Random House.

- ◇ Deming, W. E. (1986). *Out of the Crisis*. MIT Center for Advanced Engineering Study.

- ◇ Wageman, R., Nunes, D. A., Burruss, J. A. & Hackman, J. R. (2008). *Senior Leadership Teams*. Harvard Business School Press.

- ◇ Bakker, A. B. & Demerouti, E. (2007). The job demands and resources model: state of the art. *Journal of Managerial Psychology*, 22(3).

IN ONE LINE

Most of the difference in performance sits in the design of the environment, which is the part leaders control and the part they most often leave to chance.

THE PROBLEM

The high performance illusion

Many environments that describe themselves as high performance are really high output environments. They obsess over the visible result, reward short term success and pay little attention to the condition of the system producing it. None of the failures that follow looks like failure at the time. The blueprint identifies three of them, in which results are placed above people, short term wins are placed above long term health and the human cost of execution is ignored because it rarely appears in the numbers leaders review each week.

The cost arrives later and usually all at once. When capacity is ignored and interference is tolerated for long enough, the consequences become predictable. In sport they are especially visible because bodies break down and fatigue shows up in decisions made at speed. Every professional environment pays the same costs in some form, even where nobody is keeping score.

The deeper problem is that those costs are usually misattributed. Burnout gets treated as an individual resilience issue, a poor decision gets traced to the person who made it and turnover gets explained through pay or personality. Each explanation may be partly true, yet together they keep attention fixed on individuals and away from the design choices that made those outcomes more likely in the first place.

LEVER ONE

Building capacity

Capacity is the engine of sustainable performance. It makes good performance easier to produce, easier to repeat and more resilient when conditions change, which is why the blueprint treats it as something leaders build on purpose and keep investing in long after the initial effort has been recognised.

The model names five components, each of which can be strong or thin in a given environment, with the thinnest usually setting the ceiling for everything else. Each is described below alongside the signs that it is running thin and a first move that any leader can make within a week.

COMPONENT 1

People & Energy

+ Mainly additions

The physical, cognitive and emotional resources available to do the work well. Capacity cannot be separated from recovery, health, workload or the quality of the working environment, because a tired and overstretched group has less of everything else to offer.

When it is thin, staff are busy all the time yet struggle to find space for the work that matters most, sickness and minor injury cluster around demanding periods and the same people are always the last to leave.

THE COST

When capacity is ignored and interference is tolerated

- ◇ Burnout across staff and athletes
- ◇ Injuries that cluster around demanding periods
- ◇ Poor decisions made under fatigue
- ◇ High turnover among the people you can least afford to lose
- ◇ Fragile performance that holds only while nothing goes wrong

WHY IT MATTERS

Capacity makes performance easier, repeatable and resilient. The environments that invest in it consistently find that results become less dependent on any single person.

FIRST MOVE

Map staff workload against recovery with the same seriousness you apply to athletes, then protect a minimum recovery standard through the busiest weeks of the year. [Shadow Load](#) covers this in depth.

COMPONENT 2

Skills & Capability

- + Additions that multiply

This covers the technical, tactical, leadership and decision making capability each role requires. Development expands what the system can reliably ask people to do, which makes it one of the few investments that raises the ceiling on what the organisation can attempt.

When it is thin, work is repeatedly escalated to the same senior people, new staff take a long time to become useful and development conversations happen only once someone is already struggling.

FIRST MOVE

Identify the three decisions each role makes most often and build deliberate practice around them, including for staff whose own development usually stops once they are hired.

COMPONENT 3

Cohesion & Trust

- x Mainly multipliers

This is the quality of relationships that allows information to move quickly, disagreement to stay productive and people to act without unnecessary self protection.

When it is thin, problems surface late, meetings stay polite while the real conversations happen afterwards in smaller groups and people protect their own area before they help another.

FIRST MOVE

Name one recurring disagreement the group avoids and give it a structured place on the agenda, with a leader modelling how to disagree without penalty.

COMPONENT 4

Role Clarity & Processes

- x Mainly multipliers

Clear ownership, decision rights and repeatable processes reduce wasted energy and make coordinated execution far more likely. When people know what they own and how work moves between them, less capacity is lost to working out who should be doing what.

When it is thin, two people complete the same task while a third falls between them, decisions get revisited because nobody was sure they had been made and new staff learn the real process by trial and error.

FIRST MOVE

Write down who owns each of the ten most frequent decisions in your area and who needs to be consulted, then test that map against a real week of work.

COMPONENT 5

Tools & Infrastructure

- + Mainly additions

Technology, staffing, information and facilities exist to support the work, although each one can just as easily add friction when it is chosen or maintained poorly.

When it is thin, data is entered twice, reports are built by hand every week and staff keep private spreadsheets because the official system does not give them what they need.

FIRST MOVE

Ask each staff member which single tool or process costs them the most time for the least value, then fix or remove the most common answer.

Reducing interference

Interference is anything that consumes capacity without creating equivalent value. The model names seven common forms, all of which most leaders will recognise in their own environment within a few minutes of honest looking.

The filter in the blueprint matters here. Every serious performance environment contains pressure, constraint and competing demands, so any promise to eliminate interference entirely deserves suspicion. The objective is to separate the constraint that comes with the job from the waste that simply accumulates, then reduce the unnecessary portion before it reaches the people doing the work.

Interference rarely announces itself. It builds one reasonable request, one extra meeting and one unresolved tension at a time, which is why it has to be looked for deliberately and on a regular rhythm.

THE FILTER QUESTION

Is this pressure part of the job we have chosen to do, or is it waste that our own design has created? Keep the first and reduce the second before it reaches the work.

FORM AND TYPE

WHAT IT LOOKS LIKE

FIRST MOVE

Overload

– Subtraction

There are more priorities than people, everything is labelled urgent and nothing is ever removed from the list.

Keep a stop list alongside the priority list, so that every new commitment names what it replaces.

Friction

– Subtraction

Simple tasks pass through several steps, approvals and handoffs, so work that should take an hour takes a week.

Walk one routine task from start to finish and remove every step nobody can justify.

Confusion

÷ Divider

People work hard in slightly different directions because priorities, standards or roles have never been made explicit.

Write the three priorities for the next period on one page and check that every staff member would describe them the same way.

Fear

÷ Divider

People withhold bad news, avoid disagreement and choose the defensible option over the right one.

Meet the next piece of bad news with curiosity before judgement and let the group see that you did.

Ego & Politics

÷ Divider

Energy goes into status, territory and credit while decisions get shaped by who proposed them more than by their merit.

Review options before names are attached to them, so the idea can be judged apart from its author.

Misalignment

÷ Divider

Departments optimise for their own measures in ways that undercut each other, so the whole performs below the sum of its parts.

Agree one shared outcome measure that every department contributes to and review it together each month.

Decision Drag

÷ Divider

Decisions are delayed, revisited or escalated repeatedly, which leaves people waiting and capacity sitting idle.

Give anything discussed more than twice a named decision owner and a date by which the decision will be made.

Working both sides of the equation

Capacity and interference have to be considered together. Adding capability to a poorly designed system can simply give that system more resources to waste, which is what happens when an organisation hires more staff into an environment whose real constraint is unclear decision rights. Removing friction without building capability has the opposite problem, producing a calmer environment that still lacks the skill or energy to perform when it counts.

In practice every improvement cycle should include at least one change on each side, which also guards against a familiar bias in which builders keep adding programmes and fixers keep removing problems while half of the engine goes untouched.

Pressure deserves particular attention because it sits on both sides. Used well, pressure sharpens attention and builds capability for the moments that matter. Used as the default operating system, it becomes one of the largest sources of interference an organisation can generate. The blueprint's position is that pressure belongs inside the system as a tool that is applied deliberately and then released. The design question is whether your environment can tell the difference between those two uses.

PAIR EVERY CYCLE

- ◇ At least one change that builds capacity
- ◇ At least one change that removes interference
- ◇ A named owner and a review date for each

Pressure should be a tool inside the system, applied on purpose and then released.

DESIGN PHILOSOPHY

THE MECHANISM

The flywheel

The flywheel at the centre of the engine is the operating mechanism that keeps the model from becoming a static checklist. It runs through four stages in sequence and then begins again, because high performance environments have no final state and the conditions that shaped this season's design will not hold for long. The action points under each stage are deliberately small, since a flywheel gains momentum through repeated turns far more than through any single push.

FOUR SYSTEM RULES

- ◇ Make good behaviour easy
- ◇ Make bad behaviour hard
- ◇ Make success repeatable
- ◇ Reduce reliance on heroics

01 Diagnose

Start with the system before the symptom. The task is to identify where capacity is strong, where it is constrained and where interference is absorbing time, energy or attention. Good diagnosis combines data with the lived experience of the people inside the environment, because each source catches what the other misses.

ACTION POINTS

- ✓ Run the Engine Audit with the leadership group and again with the staff who do the work, then compare the two.
- ✓ Keep an interference log for two weeks, recording each time work is delayed, repeated or abandoned and what caused it.
- ✓ Use the doubling test to mark each finding as an addition, multiplier, subtraction or divider.

02 Design

Translate the diagnosis into deliberate choices about roles, processes, resources, communication and standards. Good design makes the desired behaviour easier to carry out and reduces dependence on memory, personality or individual heroics.

ACTION POINTS

- ✓ Select no more than three changes with at least one on each side of the equation, tackling dividers before adding capacity.
- ✓ For each change, write down the behaviour you expect to see and how you will know it is happening.
- ✓ Test every design against the four system rules before it goes live.

03 Implement

Put the design into the real operating environment, where theoretical quality meets schedule pressure, competing priorities and ordinary human behaviour. The test that matters at this stage is whether people can actually use what has been designed on a busy Tuesday.

ACTION POINTS

- ✓ Launch each change with the people who will use it, explaining the problem it solves.
- ✓ Give every change a named owner and a date for its first review.
- ✓ Protect the change through its first demanding week, which is when old habits are most likely to return.

04 Adapt

Treat feedback as part of the design. When something does not work as intended, the system has learned something useful about itself. Adaptation keeps the model responsive to changing people, demands and context while the underlying philosophy stays intact.

ACTION POINTS

- ✓ Review each change at thirty, sixty and ninety days, asking what shifted in behaviour, decisions and output.
- ✓ Keep what works, adjust what nearly works and stop what does not.
- ✓ Record each cycle so the organisation builds a memory of its own design decisions.

APPLICATION

Putting the engine to work

A framework that lives only on a wall changes very little. The value of the Performance Engine comes from running it on a regular rhythm with real decisions attached, until the questions it asks become part of how the leadership group thinks.

The ninety day cycle below is a practical starting point for a team, a department or a whole organisation. It follows the five steps set out in the blueprint and can be scaled up or down to suit the size of the group.

BEFORE YOU START

Agree who owns the cycle, when the weekly check in happens and where the interference log will live. A cycle without an owner stalls in its first busy week.

1	Map current capacity WEEKS ONE AND TWO	Score each of the five capacity components with the leadership group and separately with the people doing the work. The gaps between the two views are often more revealing than the scores themselves.
2	Find the largest interference WEEKS ONE AND TWO	Use the interference log to rank the seven forms by cost and mark each finding as a subtraction or a divider, then separate necessary constraint from avoidable friction before choosing where to act.
3	Choose a small number of changes WEEK THREE	Select up to three design changes with at least one on each side of the equation, starting with the largest divider. Give each one an owner, a measure and a review date.
4	Implement in the real environment WEEKS FOUR TO TEN	Run the changes through normal operating conditions and watch what shifts in behaviour, decision making and output. Hold a fifteen minute check in each week that asks only what helped, what got in the way and what needs adjusting.
5	Adapt and repeat WEEKS ELEVEN TO THIRTEEN	Review the evidence, keep what worked and feed the lessons into the next diagnosis. The second cycle is usually faster and sharper than the first because the organisation has begun to learn how it actually works.

DIAGNOSTIC TOOL

The Engine Audit

Rate each capacity component from one to five, where five means it is a genuine strength in your environment, then rate each form of interference from one to five, where five means it is costing you heavily. The symbols show whether each item usually adds, multiplies, subtracts or divides.

CAPACITY	1 THIN · 5 STRONG	INTERFERENCE	1 LIGHT · 5 HEAVY
+ People & Energy	1 2 3 4 5	- Overload	1 2 3 4 5
+ Skills & Capability	1 2 3 4 5	- Friction	1 2 3 4 5
× Cohesion & Trust	1 2 3 4 5	÷ Confusion	1 2 3 4 5
× Role Clarity & Processes	1 2 3 4 5	÷ Fear	1 2 3 4 5
+ Tools & Infrastructure	1 2 3 4 5	÷ Ego & Politics	1 2 3 4 5
Capacity total / 25		÷ Misalignment	1 2 3 4 5
		÷ Decision Drag	1 2 3 4 5
		Interference total / 35	

Reading the result. Convert each total to a percentage and subtract the interference percentage from the capacity percentage. A reading of 30 or more suggests a strong engine to protect and a reading between 10 and 29 suggests capacity is present while interference takes a real share. Below 10, the system is likely relying on effort to cover design problems. Start with the lowest scoring capacity component and the highest scoring form of interference, giving dividers priority where scores are equal.

QUESTIONS FOR THE LEADERSHIP GROUP

- Where are we relying on heroics to deliver something the system should produce routinely?
- Which form of interference would our staff name first and would we have named the same one?
- What have we added this year without removing anything to make room for it?
- Who would we struggle to replace next week and what does that tell us about our design?
- When did we last change a process because of feedback from the people using it?
- Which pressure in our environment is a deliberate tool and which has simply become the default?
- Which of our dividers are we currently trying to solve with additions?

The central test for each cycle comes straight from the blueprint and asks whether the system makes good performance easier to produce again tomorrow. An honest yes means the engine is turning, while an honest no simply gives the next cycle its starting point.

Does the system make good performance easier to produce again tomorrow?

THE CENTRAL TEST

What the model is not

The Performance Engine is sometimes misread as an argument for softer standards. It makes no case for lowering standards, avoiding pressure or removing accountability. The model separates productive challenge from unnecessary interference so that a well designed environment can hold demanding standards while reducing the waste that makes those standards harder to meet.

If anything, the model raises the standard for leaders, because asking for more effort requires very little design work while building an environment in which excellent performance becomes the normal output of ordinary weeks requires a great deal. It places responsibility for the conditions of performance with the people who hold the authority to change them.

Professional baseball offers a useful reference point. When the minor league system was reformed to remove barriers and add support for players, the intent was better development through better conditions. The Performance Engine applies the same lens one layer out to the staff and systems around the player, where the equation is just as real and far less often measured.

It is also no substitute for judgement. The model gives a leadership group a shared language and a disciplined rhythm, yet the decisions about which constraint is necessary and which pressure is useful still belong to the people closest to the work and to the leaders accountable for the outcome.

WHAT IT IS NOT

- ◇ An argument for lower standards
- ◇ A way of avoiding pressure
- ◇ A removal of accountability

WHERE IT IS MOST USEFUL

The framework suits sport, leadership teams, performance departments and wider organisations. It is most useful when a group is working hard yet still experiencing recurring overload, inconsistent execution, unclear ownership, slow decisions or dependence on a few high effort individuals.

CLOSING

The test for tomorrow *performance, by design*

Sustainable high performance comes from deliberately building the capacity of the system and reducing the interference that prevents that capacity from becoming useful performance. Extracting more effort from people indefinitely produces a peak followed by a cost, which every experienced leader has seen at least once.

The work starts small with one audit, one interference log and three design changes. It compounds each time the flywheel turns, until the environment itself becomes the reason good performance keeps happening.